Faculty of Engineering & Technology Electrical & Computer Engineering Department

Circuits and Electronics Laboratory - ENEE2103
Exp2: Circuit Laws and Theorems.

Report#1

Prepared By:
Aya Dahbour 1201738

Partners:
Amani Rabee 1201512

Amani Hmidan 1200255

Instructor: Dr. Nasser Ismail
Assistant: Eng. Bilal Ismail

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Contents

Table Of Figures

Table Of Tables

1. Abstract

Resistance, voltage, and current will be measured as part of this experiment in order to assess the KVL and KCL laws' applicability and effectiveness. Also to verify the accuracy of the voltage and current division rules. Moreover, to examine the superposition theorem's reliability. to simply identify the Norton and Thevenin equivalent circuits.

2. Theory

2.1 KVL,KCL :

Using Kirchhoff's circuit law relating to the junction rule and his closed loop rule, we can calculate and find the currents and voltages around any closed circuit providing we know the values of the electrical components within it.

We saw in the Resistors tutorial that a single equivalent resistance, (R_T) can be found when two or more resistors are connected together in either series, parallel or combinations of both, and that these circuits obey Ohm's Law.

However, sometimes in complex circuits such as bridge or T networks, we can not simply use Ohm's Law alone to find the voltages or currents circulating within the circuit. For these types of calculations we need certain rules which allow us to obtain the circuit equations and for this we can use Kirchhoff's Circuit Law.

In 1845, a German physicist, Gustav Kirchhoff developed a pair or set of rules or laws which deal with the conservation of current and energy within electrical circuits. These two rules are commonly known as: Kirchhoff's *Circuit Laws* with one of Kirchhoff's laws dealing with the current flowing around a closed circuit, Kirchhoff's Current Law, (KCL) while the other law deals with the voltage sources present in a closed circuit, Kirchhoff's Voltage Law, (KVL).

2.1.1 Kirchhoff's First Law – The Current Law, (KCL)

Kirchhoff's Current Law or KCL, states that the "total current or charge entering a junction or node is exactly equal to the charge leaving the node as it has no other place to go except to leave, as no charge is lost within the node". In other words the algebraic sum of ALL the currents entering and leaving a node must be equal to zero, $I_{(exiting)} + I_{(entering)} = 0$. This idea by Kirchhoff is commonly known as the Conservation of Charge.

Kirchhoff's Current Law:

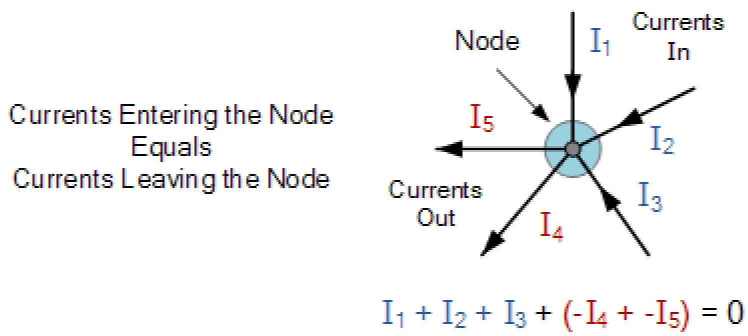


Figure2.1.1: Kirchhoff Current Law

Here, the three currents entering the node, I_1, I_2, I_3 are all positive in value and the two currents leaving the node, I_4 and I_5 are negative in value. Then this means we can also rewrite the equation as;

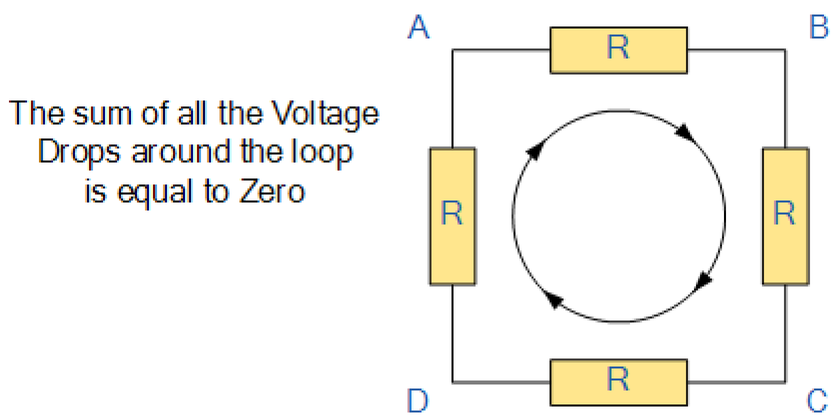
$$I_1 + I_2 + I_3 - I_4 - I_5 = 0$$

The term Node in an electrical circuit generally refers to a connection or junction of two or more current carrying paths or elements such as cables and components. Also for current to flow either in or out of a node a closed circuit path must exist. We can use Kirchhoff's current law when analyzing parallel circuits.

2.1.2 Kirchhoff's Second Law – The Voltage Law, (KVL)

Kirchhoff's Voltage Law or KVL, states that "*in any closed loop network, the total voltage around the loop is equal to the sum of all the voltage drops within the same loop*" which is also equal to zero. In other words the algebraic sum of all voltages within the loop must be equal to zero. This idea by Kirchhoff is known as the Conservation of Energy.

Kirchhoff Voltage Law



$$V_{AB} + V_{BC} + V_{CD} + V_{DA} = 0$$

Figure 2.1.2 : Kirchhoff Voltage Law

Starting at any point in the loop continue in the same direction noting the direction of all the voltage drops, either positive or negative, and returning back to the same starting point. It is important to maintain the same direction either clockwise or anti-clockwise or the final voltage sum will not be equal to zero. We can use Kirchhoff's voltage law when analyzing series circuits.

When analyzing either DC circuits or AC circuits using Kirchhoff's Circuit Laws a number of definitions and terminologies are used to describe the parts of the circuit being analyzed such as: node, paths, branches, loops and meshes. These terms are used frequently in circuit analysis so it is important to understand them.

2.2 Voltage Divider and Current Divider

Voltage Divider and Current Divider are the most common rules applied in practical electronics. As you know, there are two types of combinations in a circuit, they are series and parallel connections. Parallel circuits are also known as current divider circuits because, in these circuits, the current is divided through each resistor. Whereas, series circuits are known as voltage divider circuits because here voltage is divided across all the resistors. Voltage division rule and current division rule are necessary to understand voltage and the current flowing through each resistor. These division rules are used in most common electronic devices.

2.2.1 Voltage Divider Circuit

In order to send a current through an electric conductor, an electromotive force has to be applied. When we say the current 'I' is passing flowing through the resistor 'R,' it logically follows that a force working across resistor R. This force is known as potential difference or voltage drop across resistor R. Similarly, if we consider any part of the electric circuit three quantities i.e. voltage, current and resistance comes together.

As we got to know that, a series circuit is known as a voltage divider circuit. It is a circuit which divides the voltage into small parts. So with a power source and two resistors, we can make an easy voltage divider circuit. Here we need to connect two resistors in series combination and then apply a voltage source across the series circuit.

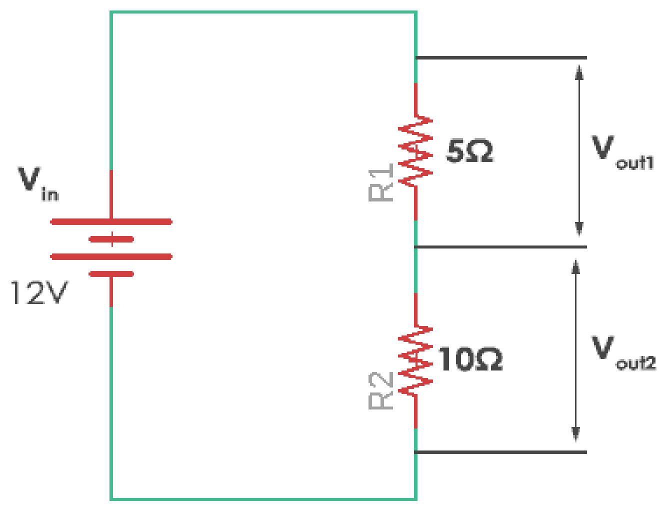


Figure 2.2.1: Voltage Divider Circuit

In this case, resistor R1 of 5 ohms and resistor R2 with 10 ohms resistance are connected. The voltages V_{out1} and V_{out2} are divided across the resistors R1 and R2. They can be calculated by a simple voltage dividing equation.

$$V_{out} = V_{in} \frac{R_x}{R_{total}}$$

Where R_x is the resistor across which we need to find the voltage and R_{total} is the total resistance (R1+R2) in the circuit. It can be simply calculated by adding all of them as they are connected in series. Thus in the given circuit, the values of the voltage across each resistor are

$$V_{out1}(voltage\ across\ R1) = vin \times \frac{R1}{R1+R2} = 12\ \times \frac{5}{5+10} = 12\frac{5}{15} = 4V$$

$$V_{out2}\ (voltage\ across\ R2) = vin \times \frac{R2}{R1+R2} = 12\ \times \frac{10}{10+5} = 12\frac{2}{3} = 8V$$

Therefore voltage across R1 is 4V and voltage across R2 is 8V. Thus here voltage is divided in the circuit across the resistors. Hence, this is called voltage divider circuit.

Voltage dividers are used in many applications but they are widely used in all types of variable resistors. Let's take an example of a potentiometer. A potentiometer is a variable resistor which can be used to create an adjustable voltage divider. The potentiometer has three terminals two of the terminals are connected to the ends of the resistor and middle terminal is connected to the wiper. It has one resistance. The two outside pins are connected to the voltage source and the middle terminal acts as a voltage divider.

2.2.2 Current Divider Circuit

A current divider is a circuit which divides the current into small parts. As we got to know that parallel circuits are current divider circuit. So with a power source and two parallel resistors, we can make an easy current divider circuit. As in the current divider network, here we need to connect two resistors in parallel combination and then apply a current source across the parallel circuit.

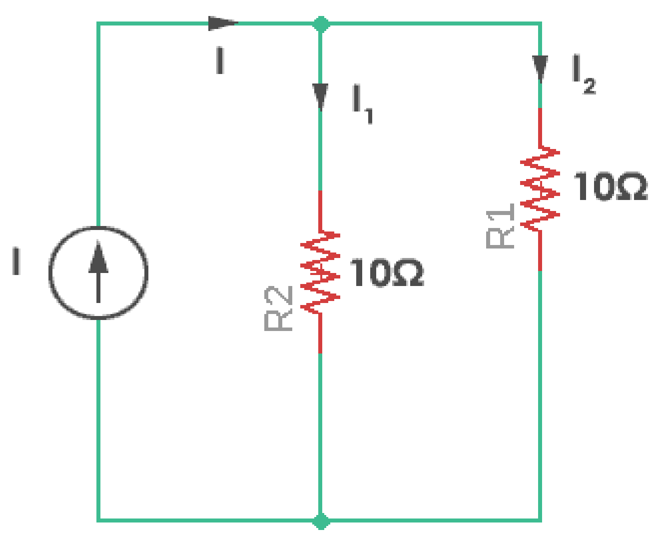


Figure 2.2.2 : Current Divider Circuit

'I_1' and 'I_2' are the current divided across resistors R1 and R2. They can be calculated by a simple current dividing equation.

$$I_n = I_{in} \frac{R_{eq}}{R_n}$$

'I_n' is the required current that flows through the resistor R_n. R_{eq} is the equivalent resistance of the parallel resistors.

Equivalent resistance (R_{eq}) is given by $\frac{1}{(\frac{1}{R1})+(\frac{1}{R2})} = 5\Omega$

So, the current flowing through resistor R1 and R2 would be

$$I_1(current\ through\ R1) = 10 \times \frac{5}{10} = 5A$$

$$I_2(current\ through\ R2) = 10 \times \frac{5}{10} = 5A$$

Here the resistors are of same value and so current will be divided in exactly half through each resistor. Thus this is known as the current divider circuit.

2.2 Superposition

Superposition theorem states that in any linear, active, bilateral network having more than one source, the response across any element is the sum of the responses obtained from each source considered separately and all other sources are replaced by their internal resistance. The superposition theorem is used to solve the network where two or more sources are present and connected.

2.2.1 Explanation of Superposition Theorem

Let us understand the superposition theorem with the help of an example. The circuit diagram is shown below consists of two voltage sources V_1 and V_2.

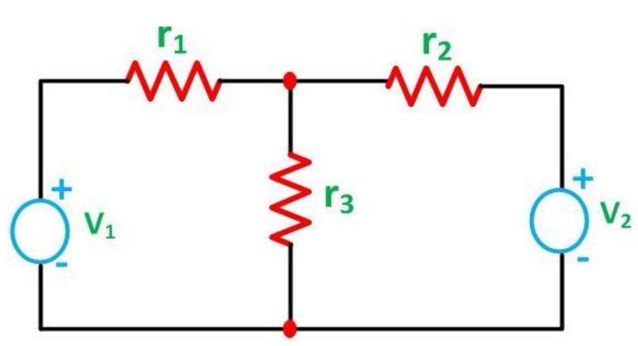


Figure 2.2.1.1 : superposition circuit

First, take the source V_1 alone and short circuit the V_2 source as shown in the circuit diagram below:

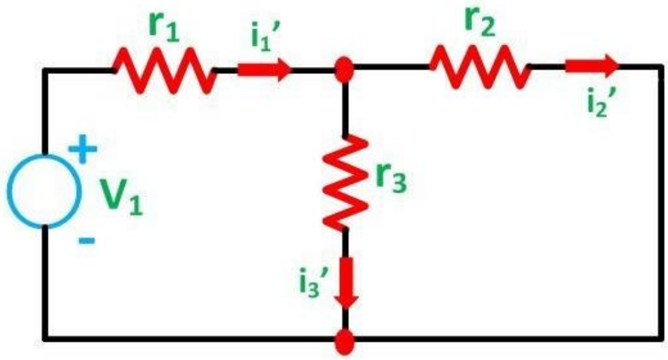


Figure 2.2.1.2 : first step in superposition

Here, the value of current flowing in each branch, i.e. i_1', i_2' and i_3' is calculated by the following equations.

$$i_1' = \frac{V_1}{\frac{r_2 r_3}{r_2 + r_3} + r_1} \dots\dots\dots\dots (1)$$

$$i_2' = i_1' \frac{r_3}{r_2 + r_3} \dots\dots\dots\dots (2)$$

The difference between the above two equations gives the value of the current i3'

$$i_3' = i_1' - i_2'$$

Now, activating the voltage source V_2 and deactivating the voltage source V_1 by short-circuiting it, find the various currents, i.e. i_1'', i_2'', i_3'' flowing in the circuit diagram shown below:

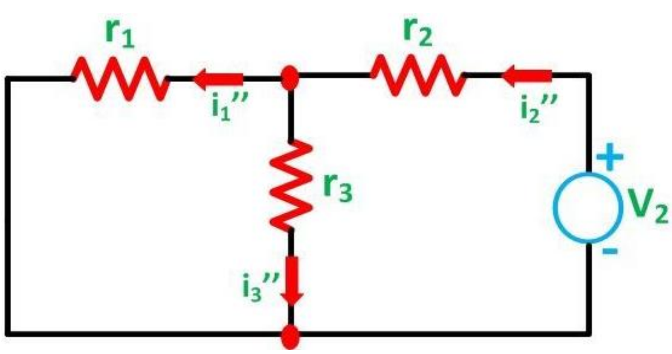


Figure 2.2.1.3 : second step in superposition

Here,

$$i_2'' = \frac{V_2}{\frac{r_1 r_3}{r_1 + r_3} + r_2} \quad \text{and} \quad i_1'' = i_2'' \frac{r_3}{r_1 + r_3}$$

And the value of the current i_3'' will be calculated by the equation shown below:

$$i_3'' = i_2'' - i_1''$$

As per the superposition theorem, the value of current i_1, i_2, i_3 is now calculated as:

$$i_3 = i_3' + i_3''$$

$$i_2 = i_2' - i_2''$$

$$i_1 = i_1' - i_1''$$

The direction of the current should be taken care of while finding the current in the various branches.

2.3 Thevenin and Norton Theorem

Thevenin's and Norton's equivalent are circuit simplification techniques that focus on terminal behavior.

2.3.1 Thevenin's Theorem

This Theorem says that any circuit with a voltage source and a network of resistors can be transformed into one voltage source and one resistor.

General Circuit Thevenin Equivalent Circuit

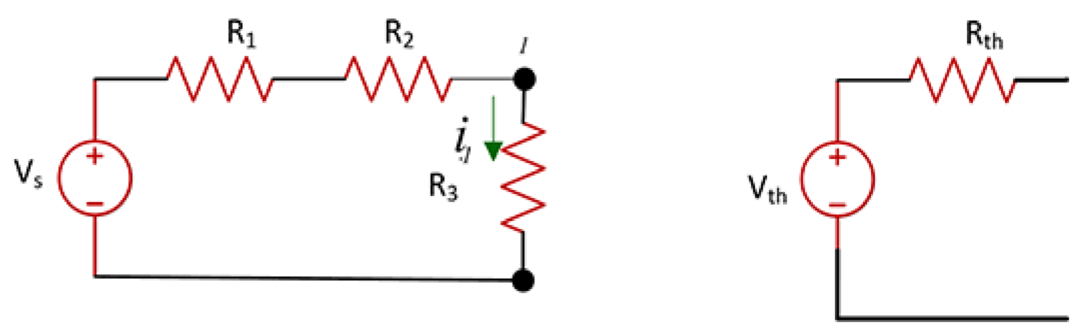


Figure 2.3.1: Thevenin circuit

Thevenin equivalent circuit represents a general circuit in a form of an independent voltage source Vth with a since resistance Rth.
To find Vth and Rth

- First, we assume that the load resistor is infinite. (Open circuit)
- Calculate Vth in original circuit using open circuit condition
- Second, reduce the load resistance to zero (short circuit). Condition, if more short circuit access the terminal and in general circuit.
- Calculate the i_{sc} (Current Short Circuit) the general circuit Isc
- The Thevenin resistance Rth can be calculated as $R_{th} = \frac{V_{th}}{i_{th}}$

2.3.2 The Norton Equivalent Circuit:

The Norton equivalent circuit represents a general circuit with an independent current source in parallel with the Norton equivalent Resistance.
Norton current source is equivalent to the short-circuit current at the terminal a and b
Norton resistance is th $i_N = i_{sc}$e same as the Thevenin resistance .

For the previous example

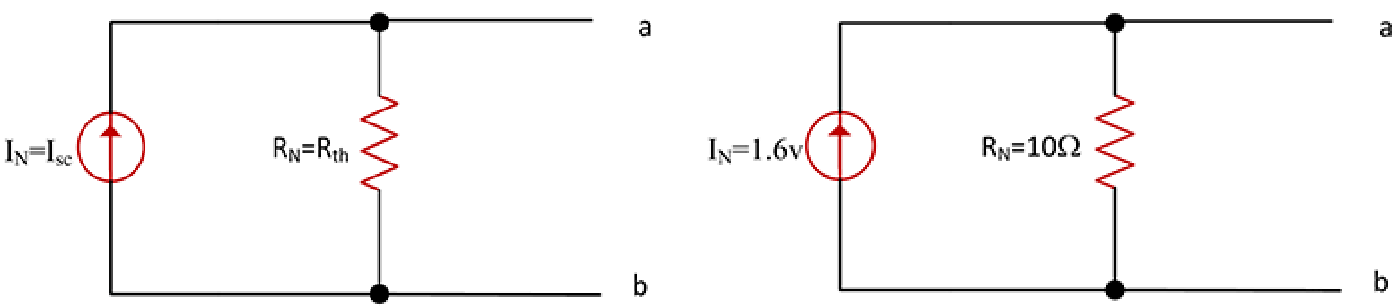


Figure 2.3.2 : Norton circuit

3. Procedure

3.1 KVL,KCL :

1.Rx=R1=R4=1kΩ , R5=3.3kΩ,R6=4.7kΩ and Vs= 15V are Connected .
2.The voltage source to 15 volts is set .
3.The resistance decade box Rx is set to the value 1kΩ.
4.The values required in table 2.1 are measured and filled.
5.The value of R_x is changed to the half of the first value.
6.Step 5 is reapeated.
7. The validity of KCL , KVL for the circuit is verified from the measurements .

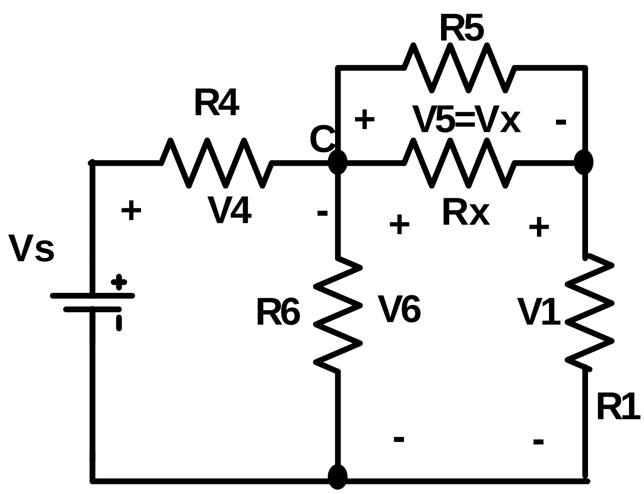


Figure 3.1: KVL,KCL circuit

3.2. Voltage & Current Division:

3.2.1 Voltage division

1. In the previous circuit the resistance R_5 is disconnect .
2. The voltages are measured on all the branches of the circuit.
3. R_x is changed as shown in table2.2 and step2 is repeated for all the given values.
4. In each case the resistors values are applied into the voltage division formula .

3.2.2 Current division

1. In the circuit R_5 is reconnected and R1 is replaced by a short circuit and R_x is set to its start value.
2. The currents are measured in all the resistive branches of the circuit and the values are filled in the table.
3. R_x is changed as shown in table 3 and step 2 is repeated for all the given values.
4. In each case the resistors values are applied into the current division formula.

3.3 Superposition:

1.The circuit of Fig 3.3 is connected.
2.The source V_{s1} is set to 5 volts and V_{s2} to 10 volts.
3.The variable resistor R_x is set to the value of 1K.
4.The current and the voltage on R_6 measure.
5.vs1 is set to zero and V_{s2} to 10 volts and the current and the voltage on R_6 is measured.
6.V_{s1} is set to 5 volts and V_{s2} to zero and the current and the voltage on R_6 is measured.

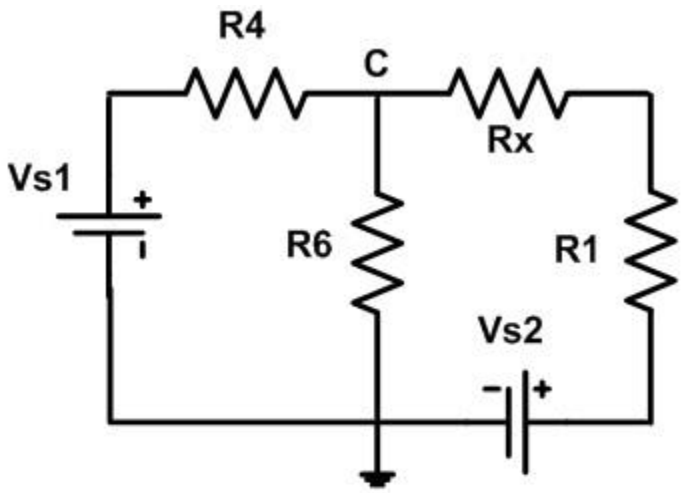


Figure 3.3: Superposition circuit

3.4 Thevenin and Norton equivalent circuits:

1. The circuit of Fig 3.3 is reconnected.
2. The V_{s1} set to 5volts and Vs2 to 10 volts and voltage across R1 is measured.
3. R_1 is disconnected and the voltage on the terminals (a,b) is measured where R_1 was connected as in Fig 3.4.[Voc- open circuit voltage]
4. The terminals (a, b) is been short circuit and the current is measured in the short circuit .
5. The voltage sources are disconnected and short circuit the terminals where each source was connected.
6. The resistance from the terminals (a,b) is measured (Rab=Rth).
7. The voltage source is connected in series with the variable resistance from the potentiometer (VR1) as in Fig 3.5 , R1 isn't .
8. The voltage source is set to Voc measured in step (3) and the variable resistance to Rth measured in step (4).
9. The voltage is measured on the opened terminals of the series connection.
10. Short circuit the terminals of the series connection and the current in the short circuit is measured.

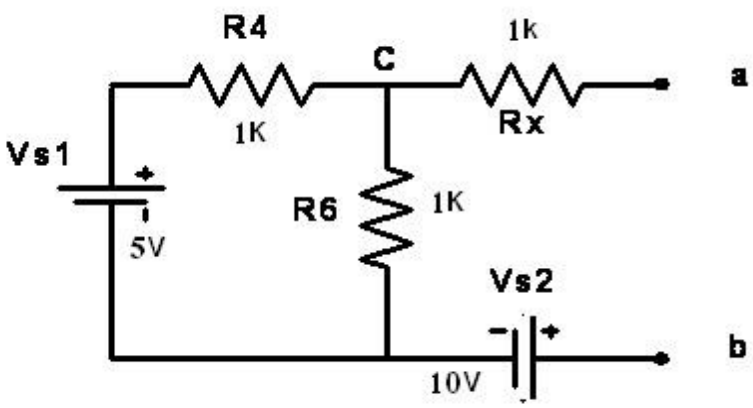


Figure 3.4: Thevenin way

VR1=Rth

Components List: R1=R4=1 kΩ , R5=3.3 kΩ, R6=4.7kΩ

4.Data & Results

4.1 KVL,KCL :

As we can see from tables (4.1)and(4.2) the findings were nearly identical, proving the validity of the Kirchhoff's laws theory that is mentioned in the theory part.

Let's test the Kirchhoff's Current Law, which says that the equivalent to the sum of current at a node is zero, for Rx = 1k, for instance, in a PSpice simulation.

6.61 mA - 1.13 mA - 1.79 mA - 3.69 mA = 0 in this situation, where I4 - I5 - I6 - Ix = 0

Let's put to the test Kirchhoff's Voltage Law, which claims that the voltages around a closed-circuit loop have a mathematical total of zero.

In this case, 15 Volts - 6.57 Volts - 8.46 Volts = 0 ,where Vs - V4 - V6 = 0.

These are the practical results for this part :

Table 4.1 : KVL,KCL practical

Vs	Pot	R1		R4		R5		R6		Rx	
		V1	I1	V4	I4	V5	I5	V6	I6	Vx	Ix
15V	Rx	4.76	4.82	6.57	6.61	3.7	1.13	8.46	1.79	3.7	3.69
15V	0.5 Rx	5.46	5.53	7.16	7.19	2.4	0.74	7.87	1.67	2.4	4.78

PSpice results :

Table 4.2: KVL,KCL PSpice

Vs		R1		R4		R5		R6		Rx	
		V1	I1	V4	I4	V5	I5	V6	I6	Vx	Ix
15V	Rx=1k	4.772	4.772	6.57	6.566	3.66	1.11	8.434	1.794	3.66	3.662

We can see that the results are almost equivalent, which supports the validity of the experiment and the theory's justification.

4.2. Voltage & Current Division:

4.2.1 Voltage division

These are the practical results for this part:

Table 4.2.1: voltage division

Vs (volt)	Pot.	V1	V4	V6	Vx
10	Rx	2.91	4.18	5.88	2.96
10	0.5Rx	3.54	4.7	5.35	1.81

PSPice results :

(For Rx=1k)

V4 = 10v – 3.975v = 6.025v

V5 = Vx = V6 = 3.975v

Data above show that the results are close, but we can also see that there is some error in the practical section because of the wires since they still have some resistivity. Even so, the results are still close enough to conclude that the theory of voltage division that we displayed in the theory section was accurate.

4.2.2 Current division

the practical results: (mA)

Table 4.2.2: current division

Vs (volt)	Pot.	I4	I5	I6	Ix
10	Rx	6.08	1.23	0.85	3.99
10	0.5Rx	7.19	0.88	0.61	5.69

PSPice results :

Table 4.2.2: Current Division PSpice

Vs (volt)	Pot.	I4	I5	I6	Ix
10	Rx	6.025mA	1.205mA	845.72uA	3.975mA

We observe that the outcomes are essentially equal, which confirms the experiment's accuracy and the theory's explanation.

4.3 Superposition:

the practical results:

Table 4.3.1: Superposition

Vs1(volt)	Vs2 (volt)	V6 (volt)	I6 (mA)
5	**10**	**5.84**	**1.24**
0	**10**	**2.93**	**0.62**
5	**0**	**2.9**	**0.61**

PSpice results :

Table 4.3.1: Superposition PSpice

Vs1(volt)	Vs2 (volt)	V6 (volt)	I6 (mA)
5	**10**	**5.938v**	**1.242mA**
0	**10**	**2.919v**	**621.12uA**
5	**0**	**2.919v**	**621.12uA**

Then, for the Superposition tables, we can see from above that the results are near, so we can build a test to prove that the technique of the superposition works.

- When Vs1 = 5 Vs2 = 10 Vs >>>> V6 = 5.84 Vs, I6 = 1.24 mA
- When Vs1 = 0 Vs2 = 10 Vs >>>> $V6_1$ = 2.93 Vs, and $I6_1$ = 0.62 mA
- When Vs1 = 5 Vs2, Vs2 = 5 Vs >>>> $V6_2$ = 2.90 Vs , and $I6_2$ = 0.61 mA
- ($V6_1$ + $V6_2$ = 2.93 + 2.9 =5.83v) & ($I6_1$ + $I6_2$ = 0.62 + 0.61 =1.23mA)
  So superposition theory has come true

Furthermore, if we run a PSpice Simulation test, the values apply to the same technique.

4.4 Thevenin and Norton equivalent circuits:

the practical results:

Vth = -1.75 v
Rth = 1.82k
I = -0.959mA

PSpice results :

Vth = 5.877 v
Rth = 1.825K
I = 2.08mA

My theoretical solution:

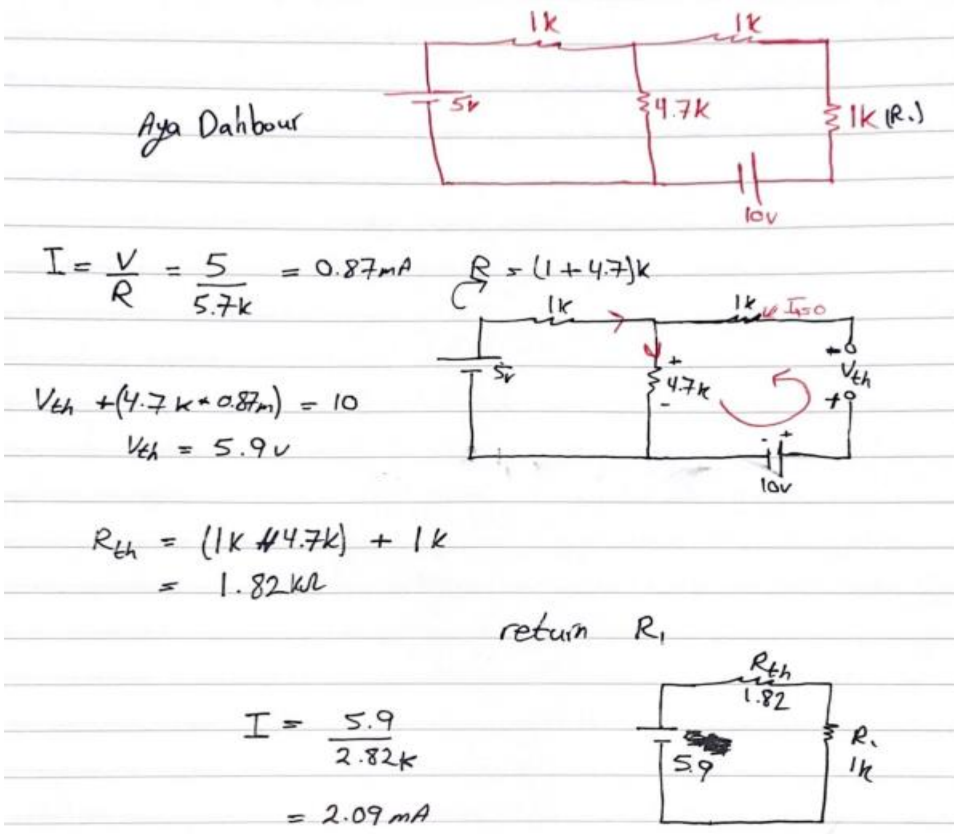


Figure 4.4: Thevenin theoretical solution

I solved it to make sure which answer is correct and which one is defective, so it turns out that the results of PSpice are correct and our practical results in the laboratory are wrong, and this could be due to several reasons, the most important of which is the lack of attention that there is an error in the installation.

5.Conclusion

Resistances, voltage, and current were measured, recorded, and compared with the exact numbers from PSpice, and all theories (KVL,KCL / Voltage & Current Division) were validated, with the exception of Thevenin, which we discovered to have a mistake that was clarified.

6. References

[1]: https://www.electronics-tutorials.ws/dccircuits/dcp_4.html

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[2]: https://www.codrey.com/dc-circuits/voltage-divider-and-current-divider/

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[3] : https://circuitglobe.com/what-is-superposition-theorem.html

Accessed on 2/11/2022 at 9:02 PM

[4]: https://www.cpp.edu/~elab/projects/project_08/index.html

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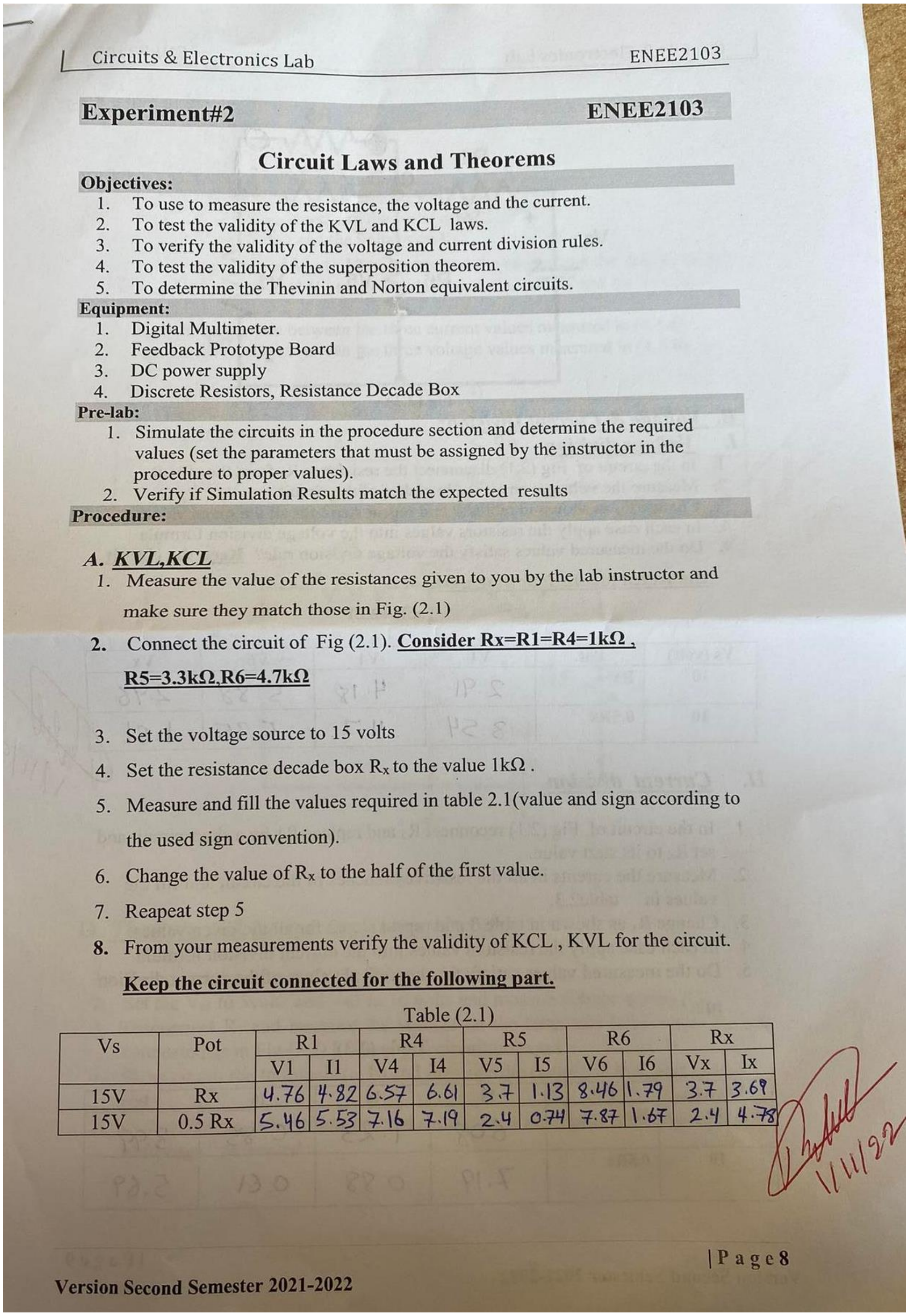
7.Data during the experiment :



Circuits & Electronics Lab ENEE2103

Experiment#2 **ENEE2103**

Circuit Laws and Theorems

Objectives:

1. To use to measure the resistance, the voltage and the current.
2. To test the validity of the KVL and KCL laws.
3. To verify the validity of the voltage and current division rules.
4. To test the validity of the superposition theorem.
5. To determine the Thevinin and Norton equivalent circuits.

Equipment:

1. Digital Multimeter.
2. Feedback Prototype Board
3. DC power supply
4. Discrete Resistors, Resistance Decade Box

Pre-lab:

1. Simulate the circuits in the procedure section and determine the required values (set the parameters that must be assigned by the instructor in the procedure to proper values).
2. Verify if Simulation Results match the expected results

Procedure:

A. <u>KVL,KCL</u>

1. Measure the value of the resistances given to you by the lab instructor and make sure they match those in Fig. (2.1)
2. Connect the circuit of Fig (2.1). **<u>Consider Rx=R1=R4=1kΩ , R5=3.3kΩ,R6=4.7kΩ</u>**
3. Set the voltage source to 15 volts
4. Set the resistance decade box R_x to the value 1kΩ .
5. Measure and fill the values required in table 2.1(value and sign according to the used sign convention).
6. Change the value of R_x to the half of the first value.
7. Reapeat step 5
8. From your measurements verify the validity of KCL , KVL for the circuit.

<u>Keep the circuit connected for the following part.</u>

Table (2.1)

Vs	Pot	R1		R4		R5		R6		Rx	
		V1	I1	V4	I4	V5	I5	V6	I6	Vx	Ix
15V	Rx	4.76	4.82	6.57	6.61	3.7	1.13	8.46	1.79	3.7	3.69
15V	0.5 Rx	5.46	5.53	7.16	7.19	2.4	0.74	7.87	1.67	2.4	4.78

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|Page 8

Version Second Semester 2021-2022

Figure 7.1: KVL,KCL data

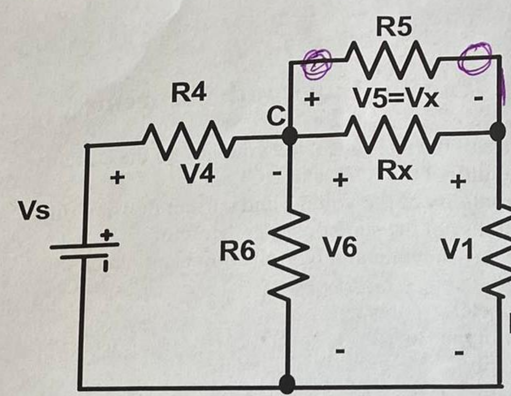


Fig (2.1)

B. Voltage & Current Division:

I. Voltage division

1. In the circuit of Fig (2.1) disconnect the resistance R_5 from the circuit
2. Measure the voltages on all the branches of the circuit.
3. Change R_x as shown in table2.2 and repeat step2 for all the given values.
4. In each case apply the resistors values into the voltage division formula
9. Do the measured values satisfy the voltage division rule? **Keep the circuit connected for the following part.**

Table (2.2)

Vs (volt)	Pot.	V1	V4	V6	Vx
10	Rx	2.91	4.18	5.88	2.96
10	0.5Rx	3.54	4.7	5.35	1.81

1/11/22

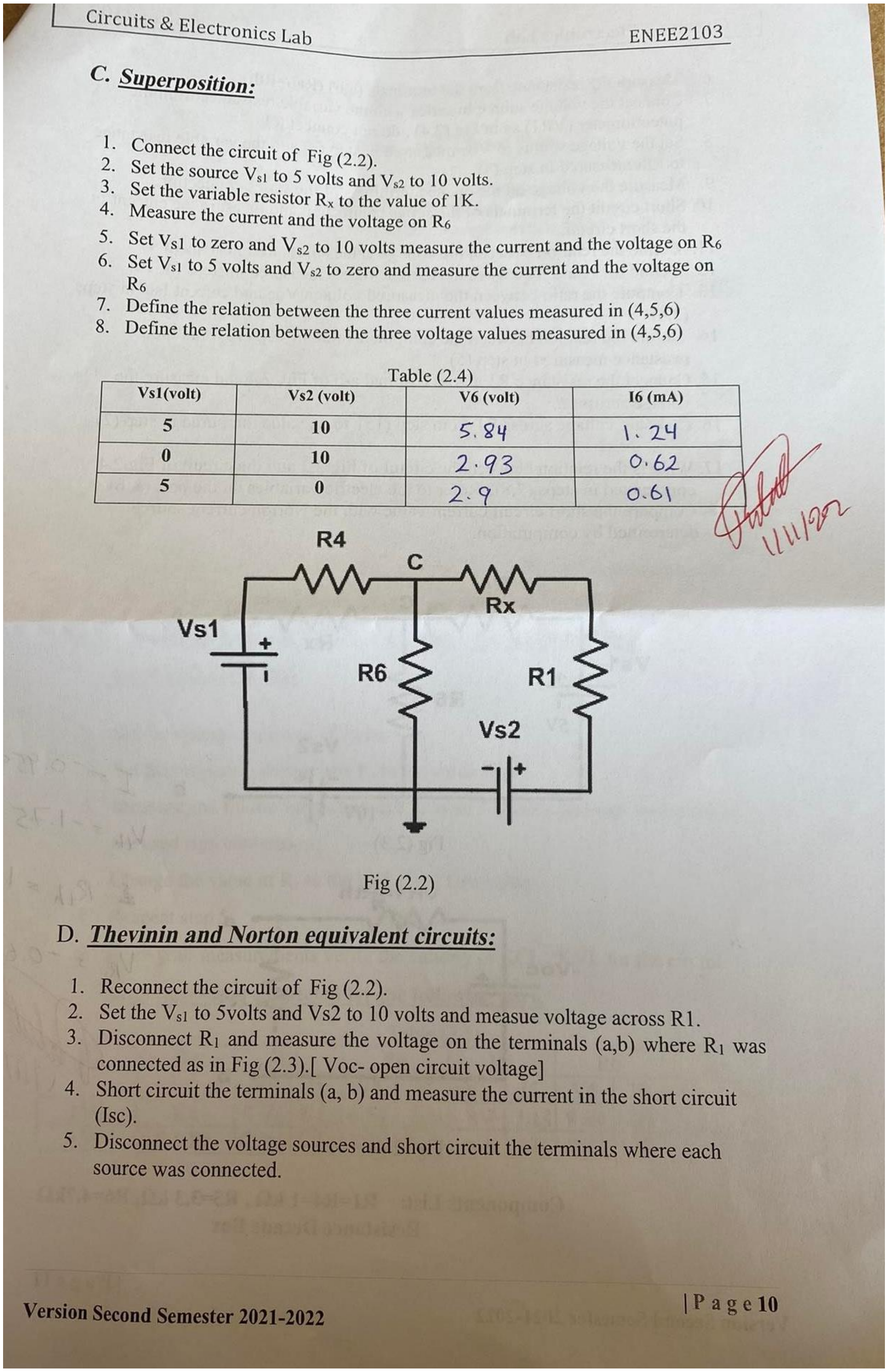
II. Current division

1. In the circuit of Fig (2.1) reconnect R_5 and replace R1 by a short circuit and set R_x to its start value.
2. Measure the curents in all the resistive branches of the circuit and fill the values in table2.3.
3. Change R_x as shown in table 3 and repeat step 2 for all the given values.
4. In each case apply the resistors values into the current division formula
5. Do the measured values satisfy the theoretical values of the current division rule?

Table (2.3)

Vs (volt)	Pot.	I4	I5	I6	Ix
10	Rx	6.08	1.23	0.85	3.99
10	0.5Rx	7.19	0.88	0.61	5.69

Figure 7.2: Voltage & Current division data



Circuits & Electronics Lab ENEE2103

C. Superposition:

1. Connect the circuit of Fig (2.2).
2. Set the source V_{s1} to 5 volts and V_{s2} to 10 volts.
3. Set the variable resistor R_x to the value of 1K.
4. Measure the current and the voltage on R_6
5. Set V_{S1} to zero and V_{s2} to 10 volts measure the current and the voltage on R_6
6. Set V_{s1} to 5 volts and V_{s2} to zero and measure the current and the voltage on R_6
7. Define the relation between the three current values measured in (4,5,6)
8. Define the relation between the three voltage values measured in (4,5,6)

Table (2.4)

Vs1(volt)	Vs2 (volt)	V6 (volt)	I6 (mA)
5	10	5.84	1.24
0	10	2.93	0.62
5	0	2.9	0.61

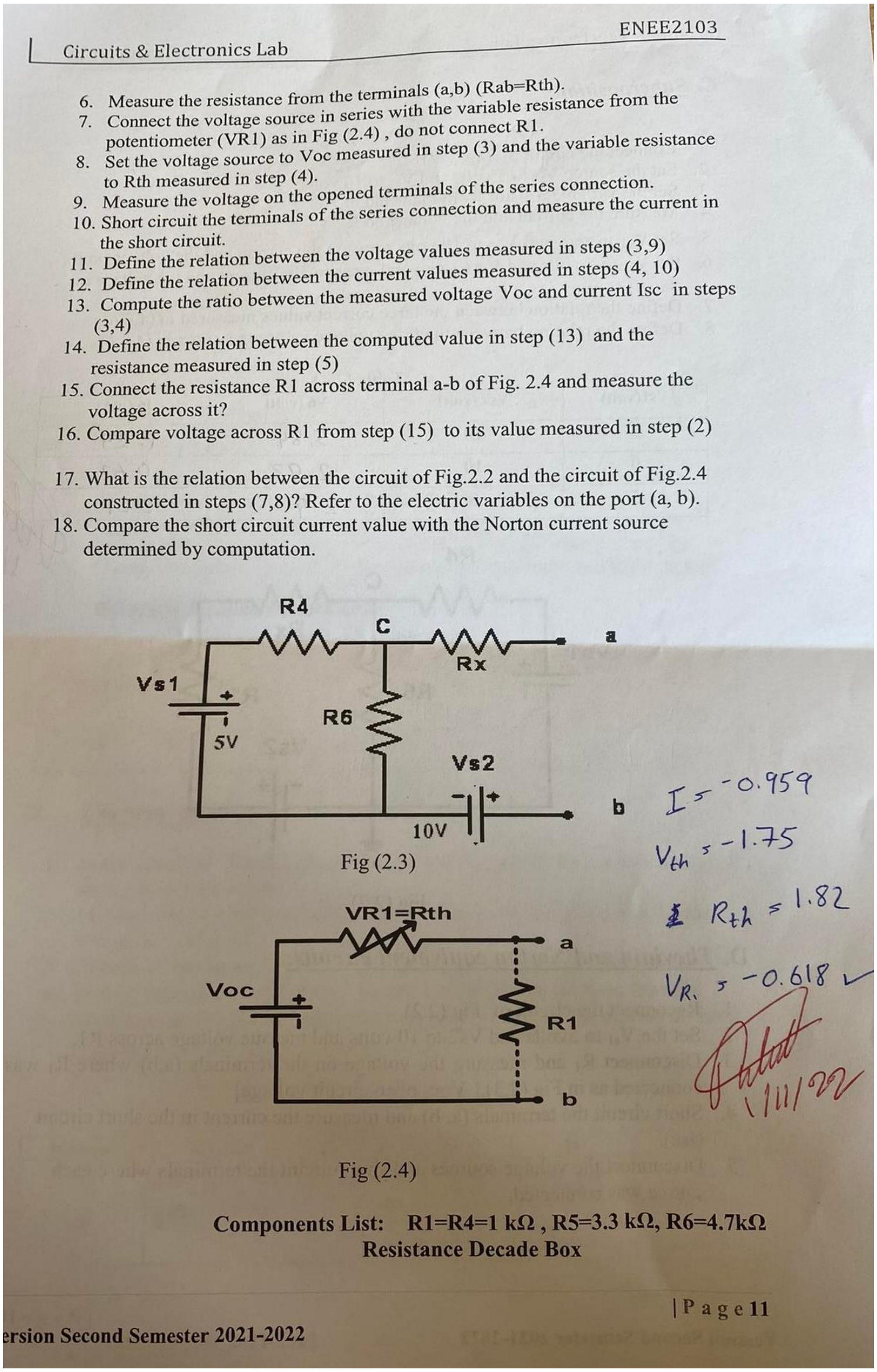
Fig (2.2)

D. ***Thevinin and Norton equivalent circuits:***

1. Reconnect the circuit of Fig (2.2).
2. Set the V_{s1} to 5volts and Vs2 to 10 volts and measue voltage across R1.
3. Disconnect R_1 and measure the voltage on the terminals (a,b) where R_1 was connected as in Fig (2.3).[Voc- open circuit voltage]
4. Short circuit the terminals (a, b) and measure the current in the short circuit (Isc).
5. Disconnect the voltage sources and short circuit the terminals where each source was connected.

Version Second Semester 2021-2022 |Page 10

Figure 7.3: Superposition data



ENEE2103

Circuits & Electronics Lab

6. Measure the resistance from the terminals (a,b) (Rab=Rth).
7. Connect the voltage source in series with the variable resistance from the potentiometer (VR1) as in Fig (2.4) , do not connect R1.
8. Set the voltage source to Voc measured in step (3) and the variable resistance to Rth measured in step (4).
9. Measure the voltage on the opened terminals of the series connection.
10. Short circuit the terminals of the series connection and measure the current in the short circuit.
11. Define the relation between the voltage values measured in steps (3,9)
12. Define the relation between the current values measured in steps (4, 10)
13. Compute the ratio between the measured voltage Voc and current Isc in steps (3,4)
14. Define the relation between the computed value in step (13) and the resistance measured in step (5)
15. Connect the resistance R1 across terminal a-b of Fig. 2.4 and measure the voltage across it?
16. Compare voltage across R1 from step (15) to its value measured in step (2)
17. What is the relation between the circuit of Fig.2.2 and the circuit of Fig.2.4 constructed in steps (7,8)? Refer to the electric variables on the port (a, b).
18. Compare the short circuit current value with the Norton current source determined by computation.

Fig (2.3)

$I = -0.959$

$V_{th} = -1.75$

$R_{th} = 1.82$

$V_{R_1} = -0.618$

1/11/22

Fig (2.4)

Components List: R1=R4=1 kΩ , R5=3.3 kΩ, R6=4.7kΩ

Resistance Decade Box

|Page 11

ersion Second Semester 2021-2022

Figure 7.4: Thevenin data